

📌 Exact exam algorithm (do this every time)

Solving AKI + urine microscopy / dipstick questions

Step 1 — Pigment vs true hematuria

- Blood+ dipstick + **no RBCs** → pigment
- Blood+ dipstick + lots of RBCs / RBC casts → glomerular bleeding

Step 2 — If pigment: rhabdo or hemolysis?

- Rhabdo clues: CK high, crush/seizure/immobilization, muscle pain/weakness
- Hemolysis clues: anemia, ↑LDH, ↓haptoglobin

Step 3 — If RBC casts present: GN

Step 4 — If WBC casts/pyuria: AIN

Step 5 — If muddy brown casts: ATN

Common traps

- "Opiate abuse" → often actually rhabdo due to immobilization
- "Blood in urine = GN" → only if **real RBCs / dysmorphic RBCs / RBC casts**
- Dipstick sees **heme**, not intact RBCs

1. Exam algorithm header

WHAT YOU SEE

Exact exam algorithm (do this every time): Solving AKI + urine microscopy / dipstick questions

WHAT IT ENCODES

Approach every dipstick/sediment vignette in a fixed order: (1) read the dipstick panel — protein, blood/heme, leukocyte esterase, nitrites, glucose, ketones, specific gravity, pH, bilirubin; (2) correlate with microscopy (RBCs/dysmorphic RBCs, WBCs, casts, crystals, epithelial cells); (3) localize by cast type. Casts are the localizing key: RBC=glomerular, WBC=AIN/pyelo, muddy brown granular=ATN, hyaline=prerenal, broad waxy=CKD, fatty/oval fat bodies (maltese cross)=nephrotic.

BOARD TRAP

WRONG answer: jumping straight to a diagnosis from one dipstick square. Discriminator: dipstick is a screen, microscopy + casts localize. Always pair them — e.g., protein-positive dipstick alone cannot tell you glomerular vs tubular vs overflow proteinuria without the sediment and a UPCR/UACR.

2. Step 1 — pigment vs hematuria

WHAT YOU SEE

Step 1 — Pigment or true hematuria

WHAT IT ENCODES

A 'blood-positive' dipstick reacts to the peroxidase activity of HEME, so it lights up for intact RBCs, free hemoglobin (intravascular hemolysis), OR myoglobin (rhabdomyolysis). The first fork is therefore: is this real hematuria or pigment? Resolve it by spinning the urine and looking — RBCs on micro = true hematuria; heme-positive dipstick with NO RBCs = pigment (myoglobin or hemoglobin).

BOARD TRAP

WRONG answer: calling any heme-positive dipstick 'hematuria.' Discriminator: dipstick detects heme, not cells. Heme+ with absent RBCs on micro = pigmenturia (rhabdo/hemolysis), not bleeding — the supernatant is also pink in hemoglobinuria but clear in hematuria once RBCs are spun down.

3. Blood+ / no RBCs = pigment

WHAT YOU SEE

Blood+ dipstick + no RBCs → pigment

WHAT IT ENCODES

Positive heme with an empty microscopy field means a heme-containing pigment is in the urine without intact red cells: myoglobin (rhabdomyolysis) or free hemoglobin (intravascular hemolysis). Both can cause pigment-nephropathy ATN via tubular obstruction, direct toxicity, and vasoconstriction; aggressive IV fluids are the cornerstone of prevention.

BOARD TRAP

WRONG answer: attributing heme+/no-RBC urine to glomerulonephritis or 'microscopic hematuria.' Discriminator: GN gives RBCs/dysmorphic RBCs/RBC casts; an empty sediment with heme positivity is pigment, so pursue CK (rhabdo) and LDH/haptoglobin/smear (hemolysis) — not a renal biopsy for GN.

4. Blood+ / RBC casts = glomerular

WHAT YOU SEE

Blood+ dipstick + lots of RBCs / RBC casts → glomerular bleeding

WHAT IT ENCODES

RBC casts and dysmorphic (acanthocytic) RBCs prove the bleeding is GLOMERULAR — red cells squeezed through a damaged glomerular basement membrane get deformed and molded into Tamm-Horsfall casts. This defines a nephritic picture: hematuria + RBC casts + variable proteinuria + HTN + AKI (e.g., IgA nephropathy, post-strep GN, ANCA/anti-GBM vasculitis, lupus nephritis).

BOARD TRAP

WRONG answer: assuming visible blood means a bladder/urologic source or stone. Discriminator: RBC casts and dysmorphic RBCs = glomerular and mandate a nephritis workup (C3/C4, ANCA, anti-GBM, ANA, ASO); isomorphic (normal-shaped) RBCs with no casts suggest a urologic source (stone, tumor, UTI) needing imaging/cystoscopy.

5. Step 2 — rhabdo vs hemolysis

WHAT YOU SEE

Step 2 — If pigment: rhabdo or hemolysis?

WHAT IT ENCODES

Once you've established pigmenturia (heme+/no RBCs), separate the two sources with labs and history. Rhabdomyolysis: massively elevated CK (often >5,000–10,000), muscle pain/weakness, dark urine, plus its electrolyte signature — hyperkalemia, hyperphosphatemia, hypocalcemia (early), and a high anion gap. Hemolysis: anemia with elevated LDH and indirect bilirubin and LOW haptoglobin, ± schistocytes.

BOARD TRAP

WRONG answer: ordering a urine myoglobin assay to confirm rhabdo. Discriminator: serum CK is the test (myoglobin clears too fast for reliable urine testing); the clue cluster of CK >5,000 + hyperK + hyperphos + hypoCa + heme+/no-RBC urine after immobilization/crush nails rhabdo over hemolysis.

6. Rhabdo clues

WHAT YOU SEE

Rhabdo clues: CK high, crush/seizure/immobilization, muscle pain/weakness

WHAT IT ENCODES

Rhabdomyolysis releases myoglobin, K+, phosphate, and CK from damaged muscle. Triggers: crush/trauma, prolonged immobilization ('found down', overdose), seizures, extreme exertion, statins (esp. with fibrates), and certain toxins/infections. Treat with early aggressive isotonic IV fluids titrated to urine output ~200–300 mL/hr; correct hyperkalemia (the life-threatening complication); urinary alkalinization/mannitol are not routinely beneficial.

BOARD TRAP

WRONG answer: blaming the urine findings on 'opiate/cocaine nephrotoxicity' or simple dehydration in a found-down patient. Discriminator: prolonged immobilization (overdose, stroke, long lie) → compression rhabdomyolysis with CK>5,000 and heme+/no-RBC urine; the drug itself is the trigger via immobility, and the mechanism of AKI is myoglobinuric ATN.

7. Hemolysis clues

WHAT YOU SEE

Hemolysis clues: anemia, ↑LDH, ↓haptoglobin

WHAT IT ENCODES

Intravascular hemolysis releases free hemoglobin that overwhelms haptoglobin and spills into urine (heme+/no RBCs). Lab triad: anemia + high LDH + LOW haptoglobin, with elevated indirect bilirubin and reticulocytosis; a peripheral smear and DAT/Coombs localize the cause (autoimmune, MAHA/TTP-HUS with schistocytes, transfusion reaction, G6PD, mechanical valve, PNH).

BOARD TRAP

WRONG answer: calling tea/cola-colored urine with anemia 'hematuria' and chasing a urologic source. Discriminator: low haptoglobin + high LDH + spherocytes/schistocytes = hemolysis; the urine is heme+ from free hemoglobin, not intact RBCs — confirm with the hemolysis panel and smear, not cystoscopy.

8. Step 3 — RBC casts = GN

WHAT YOU SEE

Step 3 — If RBC casts present: GN (glomerulonephritis)

WHAT IT ENCODES

RBC casts = glomerulonephritis (nephritic syndrome). The pattern is hematuria + dysmorphic RBCs/RBC casts + subnephrotic proteinuria + HTN + rising creatinine. Workup is driven by complement and serologies: low complement (post-strep GN, lupus, MPGN, cryoglobulinemia, endocarditis) vs normal complement, pauci-immune (ANCA vasculitis), anti-GBM (Goodpasture), IgA nephropathy.

BOARD TRAP

WRONG answer: confusing nephritic (RBC casts, mild proteinuria) with nephrotic (>3.5 g/day proteinuria, edema, hypoalbuminemia, hyperlipidemia, oval fat bodies/maltese-cross lipiduria, bland sediment). Discriminator: RBC casts/active sediment = nephritic GN; heavy proteinuria with FATTY casts and no RBC casts = nephrotic — different differentials and workup.

9. Step 4 — WBC casts = AIN

WHAT YOU SEE

Step 4 — If WBC casts/pyuria: AIN (acute interstitial nephritis)

WHAT IT ENCODES

WBC casts with sterile pyuria and white cell casts point to acute interstitial nephritis — a hypersensitivity reaction in the interstitium. Classic causes: drugs (NSAIDs, beta-lactams/penicillins, cephalosporins, sulfonamides, PPIs, rifampin), infections, and autoimmune disease. Classic (but only ~10–30% present) triad: fever, rash, eosinophilia, often with eosinophiluria. Treat by stopping the offending drug; steroids if no recovery.

BOARD TRAP

WRONG answer: treating WBC casts/pyuria reflexively as a bladder UTI and prescribing antibiotics. Discriminator: AIN gives sterile pyuria + WBC casts + drug exposure ± eosinophiluria with a NEGATIVE culture; pyelonephritis gives WBC casts WITH bacteriuria/positive nitrites/leukocyte esterase and systemic infection — context (offending drug vs fever/flank pain + positive culture) separates them.

10. Step 5 — muddy brown casts = ATN

WHAT YOU SEE

Step 5 — If muddy brown casts: ATN (acute tubular necrosis)

WHAT IT ENCODES

Muddy brown granular casts and renal tubular epithelial cell casts = acute tubular necrosis (ischemic or nephrotoxic). Supporting labs: FENa >2% (FEurea >50% if on diuretics), urine osmolality ~300 (isosthenuria). Causes: prolonged hypoperfusion/sepsis (ischemic) and contrast, aminoglycosides, amphotericin, cisplatin, pigments (nephrotoxic). Management is supportive with dialysis for refractory hyperK, acidosis, volume overload, or uremia.

BOARD TRAP

WRONG answer: calling muddy brown granular casts 'prerenal' and giving more fluids expecting full reversal. Discriminator: prerenal = BLAND sediment/hyaline casts, FENa<1%, BUN/Cr>20, urine osm>500; established ATN shows muddy brown granular casts, FENa>2%, isosthenuria — it will not correct with a fluid bolus the way prerenal does.

11. Trap — 'opiate abuse' is rhabdo

WHAT YOU SEE

'Opiate abuse' → often actually rhabdo due to immobilization

WHAT IT ENCODES

The unresponsive/'found down' patient (opioid or other overdose, stroke, alcohol) lies immobile for hours, causing pressure-induced compression rhabdomyolysis → myoglobinuric ATN. Expect CK >5,000, heme+/no-RBC urine, hyperkalemia, hyperphosphatemia, hypocalcemia, and an anion-gap acidosis. Early aggressive IV fluids are key.

BOARD TRAP

WRONG answer: attributing the AKI to direct opioid nephrotoxicity, contrast, or 'just dehydration.' Discriminator: the mechanism is immobilization → rhabdo → pigment ATN; the giveaway is a sky-high CK with heme-positive but RBC-negative urine. Treat the rhabdo, not a phantom direct drug toxicity.

12. Trap — 'blood in urine' = GN only if real RBCs

WHAT YOU SEE

'Blood in urine = GN' only if real RBCs / dysmorphic RBCs / RBC casts

WHAT IT ENCODES

Do not equate dipstick heme positivity with glomerulonephritis. GN requires evidence of glomerular bleeding: dysmorphic (acanthocytic) RBCs and/or RBC casts on microscopy. Isolated heme positivity could be pigment (rhabdo/hemolysis), and isomorphic RBCs without casts suggest a urologic source needing imaging/cystoscopy rather than a nephritis workup.

BOARD TRAP

WRONG answer: launching a GN serology workup (or biopsy) off a positive dipstick alone. Discriminator: confirm with microscopy first — only dysmorphic RBCs/RBC casts justify the nephritic workup; isomorphic RBCs → urologic source; heme+/no RBCs → pigment workup (CK, LDH/haptoglobin).

13. Trap — dipstick sees heme not RBCs

WHAT YOU SEE

Dipstick sees heme, not intact RBCs

WHAT IT ENCODES

Core principle: the blood pad reacts to heme peroxidase activity, so it cannot distinguish RBCs from hemoglobin or myoglobin — microscopy is mandatory to interpret it. A parallel blind spot exists for protein: the dipstick protein pad detects mainly ALBUMIN and is poorly sensitive to immunoglobulin light chains (Bence-Jones protein).

BOARD TRAP

WRONG answer in myeloma vignettes: 'rule out significant proteinuria because the dipstick protein is negative/trace.' Discriminator: in multiple myeloma, heavy light-chain (Bence-Jones) proteinuria can give a NEGATIVE/trace dipstick because the pad reads albumin, not light chains — order a 24-hour urine protein, urine protein electrophoresis (UPEP), and serum/urine immunofixation to catch it. The dipstick blood pad likewise misses nothing about heme source, so always confirm with micro.
